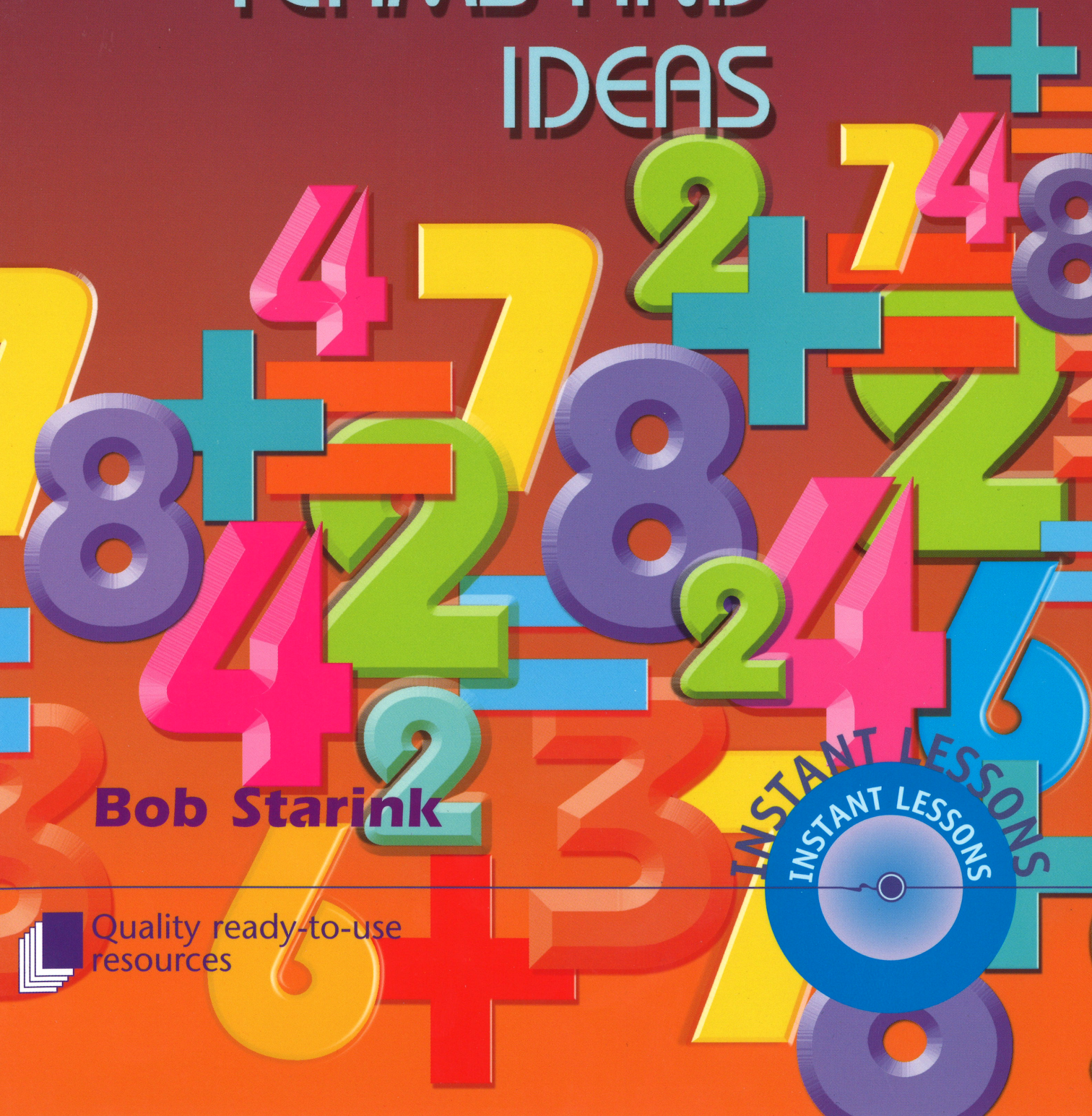
LITERACY IN MATHS

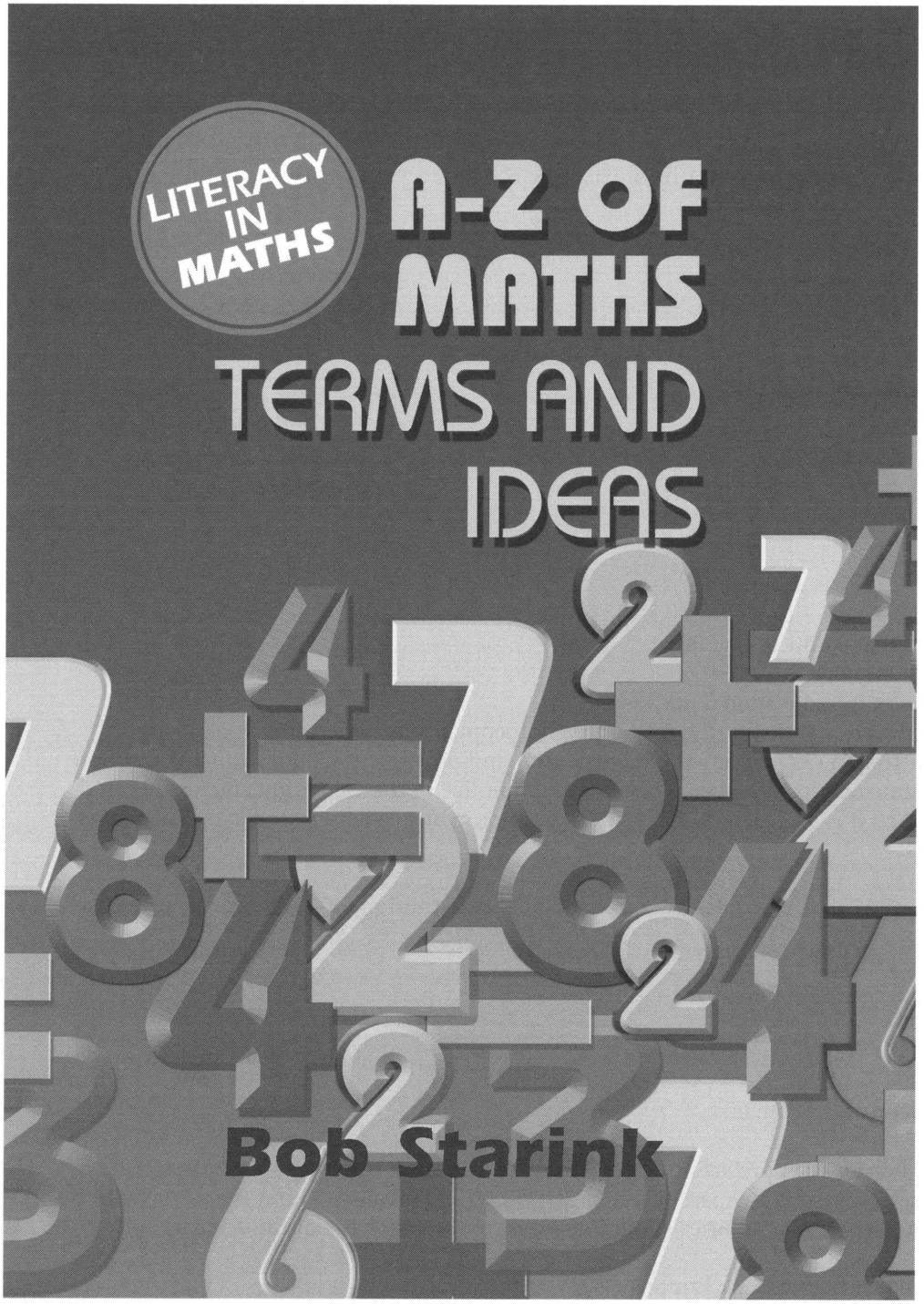
A-Z OF MATHS TERMS AND IDEAS



Bob Starink

INSTANT LESSONS

Quality ready-to-use resources



This book was commissioned by Jennifer Gough, production was by Piper Print Media and Gail McManus Graphics designed the cover.

Bob Starink has been teaching Mathematics for over 25 years at various secondary schools in Western Sydney, NSW. He collaborated with his wife Jennifer, an English teacher, to produce this *Literacy in Maths* series. Bob has co-authored other secondary mathematics textbooks, covering the full range of ability levels. He has been involved in leadership groups and has presented at curriculum workshops. Bob hopes that his *Literacy in Maths* books will improve students' coneptual understanding and make mathematics more than a repetition of processes.

© Bob Starink and Blake Education 2008

Downloadables sold with a licence to print and photocopy
The contents of this book are covered by copyright laws in Australia and in other countries.
The purchase price includes the right for you to print and photocopy these pages for your students without further payment to the publisher. For further information about how schools can use this photocopiable resource, please refer to the 'FAQ's on www.blake.com.au.
Otherwise, you may only download, print, copy, store and communicate the contents of this book without permission when that is allowed by the copyright laws applicable to you.
If you work in an Australian school that is covered by the education statutory licence, you may be allowed to copy, download, print, store and communicate parts of this book without permission from copyright owners. Under the education statutory licence, payments are made on behalf of schools to Copyright Agency, which distributes the payments to writers, artists and publishers. See www.copyright.com.au.

Copyright Agency
Level 12, 66 Goulburn Street
Sydney NSW 2000
Telephone: +612 9394 7600
Facsimile: +612 9394 7601
E-mail: memberservices@copyright.com.au

Printed in Australia by Ligare Book Printer

Blake EDUCATION
Better ways to learn

Instant Lessons
An imprint of Blake Education Pty Ltd
ABN 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au

ISBN **978 1 92571 008 3**

INTRODUCTION

Understanding the language of maths is critical to understanding mathematical concepts. This literacy resource explores the use of mathematical concepts and calculations across the letters of the alphabet. Each topic contains a list of words which are featured in a range of literacy activities for students of all ability levels. The activities provide an emphasis on interpreting, analysing, calculating and appreciating how mathematics is integral to all areas of life.

CONTENTS

1 A 4
2 B 6
3 C 8
4 D 10
5 E 12
6 F 14
7 G 16
8 H 18
9 I 20
10 J, K, X, Y, Z 22
11 L 24
12 M 26
13 N, Q 28
14 O 30
15 P 32
16 R 34
17 S 36
18 T 38
19 U, V, W 40
20 Symbols 42
21 Maths alphabets 44
Answers 46

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

1A

WORD LIST:

- angle
- apply
- appropriate
- acute
- arm
- arc
- adjacent
- alternate
- addition
- analogue
- accuracy
- abbreviation
- area
- approximate
- average
- arithmetic
- algorithm
- analyse
- axis
- axes
- algebra
- associative
- ascending

Use the word list above to complete the following exercises:

PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

	Noun (naming word)	Verb (doing word)	Adjective (describing word)
1			approximate
2		apply	
3	addition		
4			ascending
5		analyse	
6			associative
7	abbreviation		

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 angle_____

2 axes_____

3 algorithm_____

4 algebra_____

5 arm_____

6 analogue_____

7 addition_____

8 average_____

9 area_____

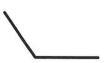
© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

10 ascending_____

11 accuracy_____

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	alternate	+	• opposite
2	analogue		• obtuse
3	adjacent	¼ past	• mean
4	arm	$P = 4L$	• exact
5	acute		• corresponding
6	addition	A 60% B 30% C 10%	• rule
7	approximate		• interpret
8	average		• pattern
9	associative	≈	• digital
10	algorithm	$(6 + 8) \div 2$	• plane
11	arithmetic	6p	• subtraction
12	analyse	7×4	• vertex
13	area		• perimeter
14	axis	$4 \times (3 \times 5) = (4 \times 3) \times 5$	• distributive
15	algebra		• operations

WHICH WORD ...

1 is an anagram of car?_____

2 ends with wood?_____

3 starts with a donkey?_____

4 has a tear in its centre? _____

5 ends angrily?_____

6 ends with a friend?_____

7 ends with a bell?_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

2 B

WORD LIST:

- base
- button
- brackets
- braces
- bit
- byte
- binary
- balance
- backtracking
- biased
- bar
- breadth
- boundary
- blocks
- build
- ball
- bisect
- billion

Use the word list above to complete the following exercises:

DOUBLE MEANING – Write two sentences using each word, one with a mathematical context and the other with a non-mathematical context.

1 balance_____

_____

2 base_____

_____

3 bit_____

_____

4 brackets_____

_____

5 bar_____

_____

6 braces_____

_____

WORDS IN A WORD – Find 42–48 words in the word BREADTH. Each word you find must have at least three letters, no 'er' words such as 'hater' and each letter can only be used as often as it appears in 'breadth'. No names are allowed!

_____

_____

_____

HALF THE ALPHABET – Change the bold letter to form a different word. You must use a different letter for each one.

1 **b**ase_____

2 **b**utton_____

3 bracke**t**s_____

4 brac**e**s_____

5 by**t**e_____

6 **b**all_____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

7 bit_____

8 billion_____

9 build_____

10 balance_____

11 bar_____

12 biased_____

13 blocks_____

MEANING MATCH – Draw lines to match each word on the left with the most appropriate meaning on the right.

1 byte	• unfair
2 bit	• decimal
3 base	• reversing
4 bisect	• sphere
5 ball	• binary digit
6 backtracking	• key
7 button	• column
8 biased	• width
9 bar	• edge
10 build	• 8 bits
11 boundary	• construct
12 breadth	• cut in half

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 biased_____

2 bar_____

3 brackets_____

4 balance_____

5 boundary_____

6 backtracking_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

3 C

WORD LIST:

- complementary
- coefficient
- cone
- cardinal
- circle
- chord
- circumference
- calculator
- common
- congruent
- coordinates
- cube
- cylinder
- cancelling
- capacity
- caret
- cubic
- Cartesian
- centre
- categorical
- character
- counting
- concurrent
- collinear
- combination
- curve
- cumulative
- construct
- constant
- class
- classification
- closed
- concave
- convex
- composite
- compasses
- commutative
- centimetre
- cross-section
- cluster
- census
- collecting
- century
- column
- convert
- certain

Use the word list above to complete the following exercises:

'A DOZEN PLEASE' – Many of the C words come from the space and geometry branch of mathematics. List any 12 of them and draw a sketch of each one.

1	2	3	4
5	6	7	8
9	10	11	12

TRIOS – Circle the correct term to complete each sentence.

1 Cubic units are used in **perimeter / area / volume.**

2 A grouping of scores is called a **class / constant / combination.**

3 A cardinal number is also called a **counting / constant / common** number.

4 The numerical part of an algebraic term is called a **character / composite / coefficient.**

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

5 The caret symbol ^ is also called **cap / hat / top.**

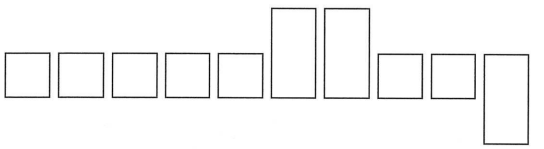
6 Cancelling is a process used with **decimals / percentages / fractions.**

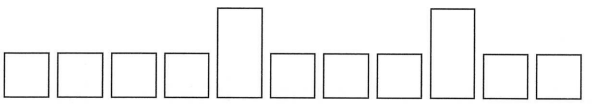
7 A shape with any diagonal inside the figure is called **concave / convex / congruent.**

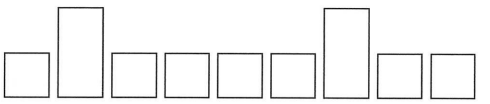
8 A centimetre is **1/10 / 1/100 / 1/1000** of a metre.

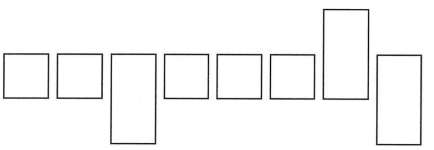
9 *Cluster, census, column* and *collecting* are all words used in **statistics / measurement / algebra.**

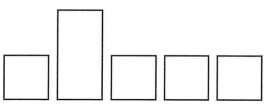
WORD SHAPES – Write list words to fit the word shapes.

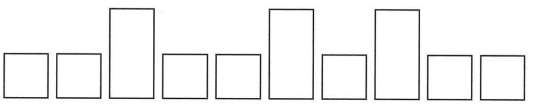
1 

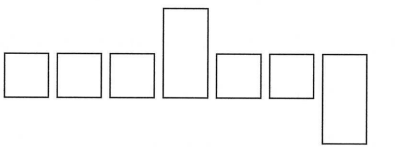
2 

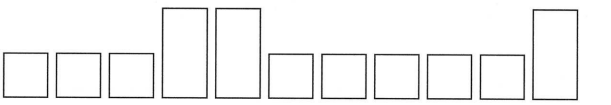
3 

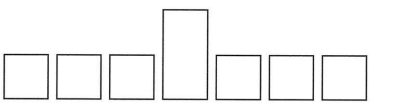
4 

5 

6 

7 

8 

9 

WHICH WORD ...

1 *doesn't win in its centre?* _____

2 *starts with a young bear?* _____

3 *ends with a grotto?* _____

4 *ends with a girl?* _____

5 *starts with a vehicle?* _____

6 *contains a room in a jail?* _____

7 *ends with electricity?* _____

8 *has a boy's name inside?* _____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

4 D

WORD LIST:

- data
- decagon
- decimal
- degrees
- denominator
- diagonal
- diameter
- difference
- digit
- dimensions
- distributive
- dividend
- division
- dodecagon
- dodecahedron
- day
- decade
- database
- document
- definition
- depth
- directed
- discrete
- diamond
- distance
- dozen
- digital
- draw
- descending
- diagram
- discount
- describe
- decrease
- display
- divided bar
- distribution
- divisor

Use the word list above to complete the following exercises:

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 draw_____

2 decimal_____

3 decagon_____

4 diameter_____

5 decade_____

6 discrete_____

7 discount_____

8 distribution_____

9 distance_____

10 data_____

SENTENCE JUMBLE – Reorder the words so that each sentence makes sense. The first word is capitalised to help you get started.

1 discount a price A is decrease in

_____

2 numbers are Integers directed

_____

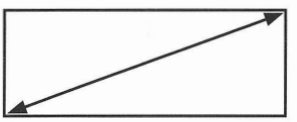
© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

3 time 12 time is hour Digital

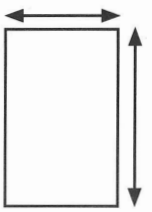
4 A description accurate definition is an

5 diamond A described be can rhombus as a

SYMBOLIC – Use list words to label the following.

1 

2 4

3 

4 $\frac{5}{8}$ ←

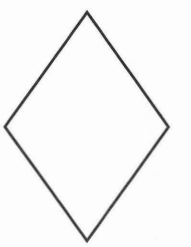
5 10, 6, 3, 0 →

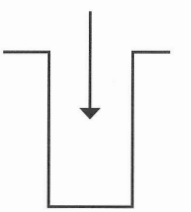
6

7 12

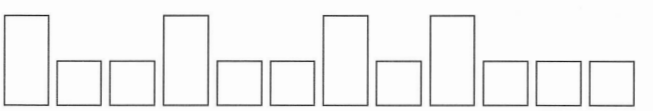
8 42 ÷ 6 ↑ ↑

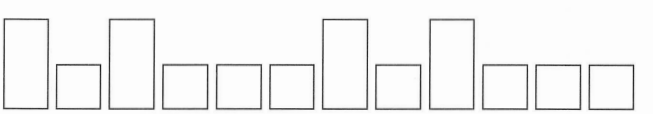
9 42° ←

10 

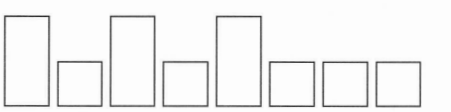
11 

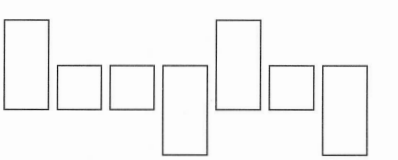
WORD SHAPES – Write list words to fit the word shapes.

1 

2 

3

4 

5 

6

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

5 E

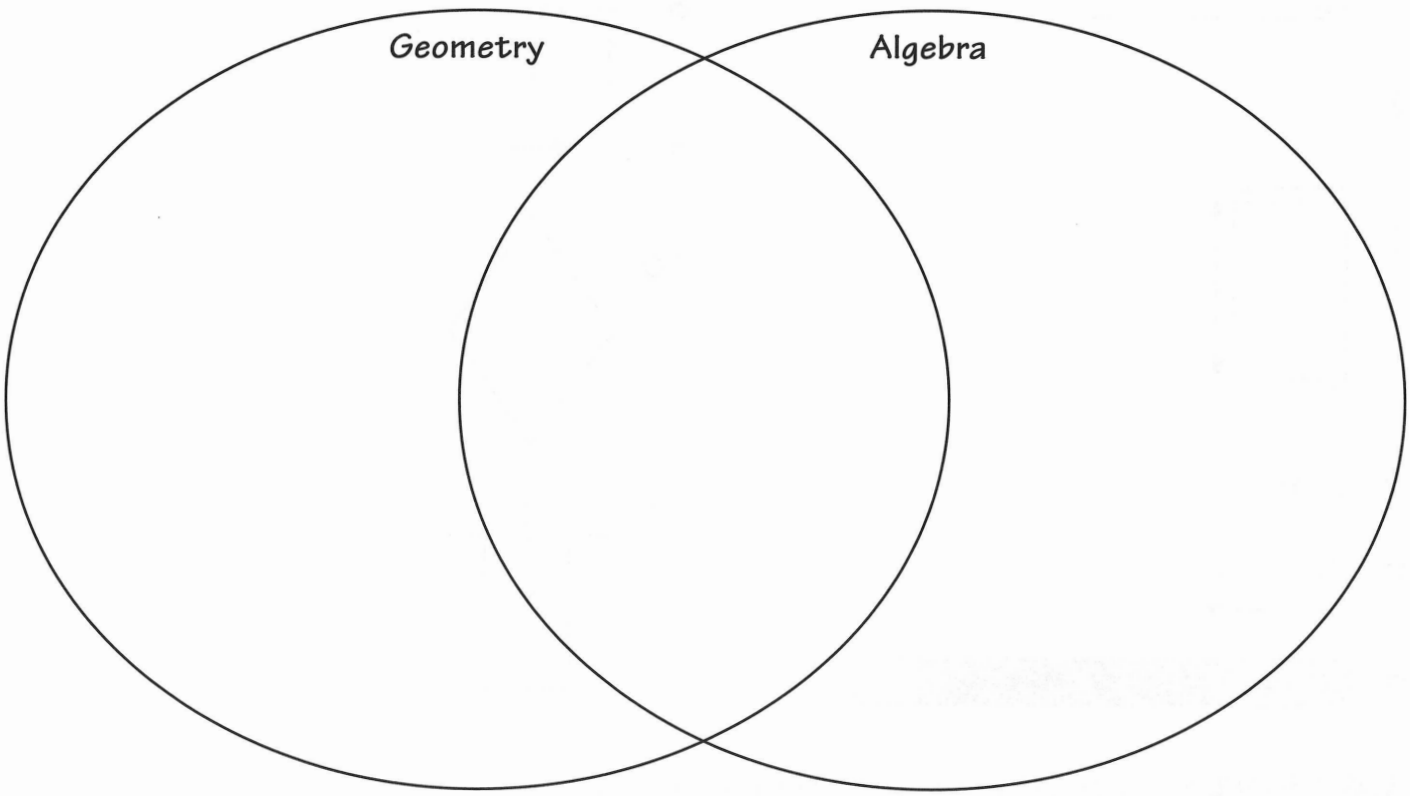
WORD LIST:

- ellipse
- equation
- equal
- equiangular
- estimate
- even
- exponent
- Euler
- Euclid
- edge
- error
- equivalent
- equilateral
- exact
- expand
- expression
- evaluate
- enlargement
- exterior
- experiment
- event
- eighth
- extend
- explain

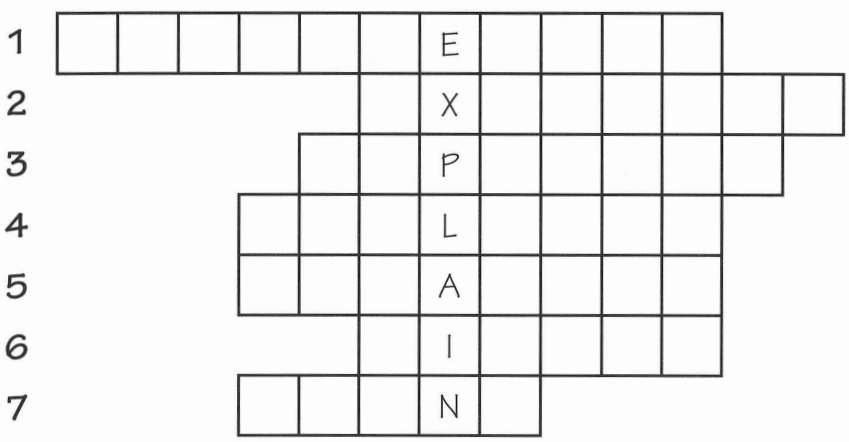
Use the word list above to complete the following exercises:

IT'S A VENN – Write the following words into the Venn Diagram, so that words associated with geometry are entirely in the geometry oval, words associated with algebra are entirely in the algebra oval, and words that can be used in both are in the intersecting section of the ovals.

Words to use: ellipse, equation, equal, equiangular, exponent, Euler, Euclid, edge, equivalent, expand, expression, evaluate, enlargement, exterior.



FILL THE GAPS – Write list words to fit each horizontal row.



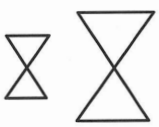
© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

	Noun (naming word)	Verb (doing word)	Adjective (describing word)
1		estimate	
2	enlargement		
3		expand	
4	equation		
5		explain	
6	experiment		
7	expression		

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	even		• reduction
2	estimate	±	• mistake
3	exterior		• equiangular
4	enlargement	2, 4, 6...	• lengthen
5	extend		• oval
6	equal	=	• index
7	equilateral	$6(p + 2)$	• equivalent
8	expand	u^4	• odd
9	ellipse		• factorise
10	exponent	$49 \times 20 \approx 1000$	• evaluate
11	error		• interior

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

6 F

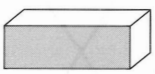
WORD LIST:

- factor
- factorise
- Fibonacci
- figurate
- finite
- fraction
- frequency
- formula
- fortnight
- face
- false
- flat
- fractal
- fair
- function
- fifth
- finance
- fifty-fifty

Use the word list above to complete the following exercises:

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	finite	$P = 4L$	• money
2	formula	fifty-fifty	• product
3	frequency	$	• infinite
4	finance	$\frac{1}{5}$	• side
5	false	———	• level
6	fair	$(x + 6)$	• rule
7	face	5, 8, 13, 21	• five
8	fifth		• triangular
9	flat	1, 3, 6, 10	• sequence
10	Fibonacci	f	• number of
11	figurate	∞	• true
12	factor	$7 = 8$	• biased

NON-MATHS – The passage below uses list words in a non-mathematical context. Use these list words to complete the text: *finite, factor, Fibonacci, frequency, fortnight, function, face, flat, false, fraction, fair.*

I have been attending the lectures of Doctor _____ with increasing _____. It used to be once a _____, now it's once a week.

Today I was in a hurry. I had just returned from a _____ after work and I was running late. Then I got stuck in traffic. I had a _____ number of

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

options. I could sit in the traffic or take a short cut. I turned into the side street, sure I'd get to the lecture in a _____ of the time. This ended up being a _____ assumption. I had forgotten to _____ in the rough road and wound up with a _____ tyre. This wasn't _____. I had to _____ the truth. I was not going to make it in time.

PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

	Noun (naming word)	***Verb (doing word)***	***Adjective (describing word)***
1		finance	
2	frequency		
3			false
4		factorise	
5	face		
6			flat
7	formula		

EVERY THIRD LETTER – Work out the hidden sentence by writing every third letter in the following text.

B D A W F F W A R G O A T H C R U T F M A T I L E D H E A A C V S S W A J P L Y U L B T I K
G T D E S U T P W E A F Y R E A T N M S E A S H T I K F M D E I A O L O P A E C R H T T D O O
R F I S C T E T S F R O D Q R T H I X A G O P I V B N A E A H R L D C S U I H O O A R E P V N E

_____

_____

FIXES – Write different words (with at least two more letters than given) which begin or end with the following prefixes (on the left) and suffixes (on the right) from list words. You cannot use list words!

1 fact_____ 2 ance_____

3 frac_____ 4 ight_____

5 fre_____ 6 ency_____

7 fun_____ 8 tal_____

9 fal_____ 10 raction_____

11 fin_____ 12 rise_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

7 G

WORD LIST:

- geometry
- gradient
- graph
- greater than
- grouping
- greatest
- gram
- gigabyte
- globe
- grid
- gross
- generalised

Use the word list above to complete the following exercises:

TRIOS – Circle the correct term to complete each sentence.

1 A gigabyte is **1 000 / 1 000 000 / 1 000 000 000** bytes.

2 A gradient is a **line / slope / polygon.**

3 The symbol for greater than is **> / < / ≥**.

4 **Brackets / Percentages / Asterisks** are an example of grouping symbols.

5 Gross pay is **after tax / after Medicare Levy / before tax** pay.

6 Generalised arithmetic is mostly used in **geometry / number / algebra.**

WRITE ...

1 three synonyms for 'greatest'._____

2 three units larger than a gram._____

3 two synonyms for 'grid'._____

4 three units smaller than a gigabyte._____

5 the simple definition of 'gradient'._____

6 four types of graph._____

7 two other meanings of 'gross'._____

WHICH WORD ...

1 is mostly a boy's name?_____

2 ends with an exam?_____

3 has a type of music in its centre?_____

4 consumes in its centre?_____

5 ends with a sheep?_____

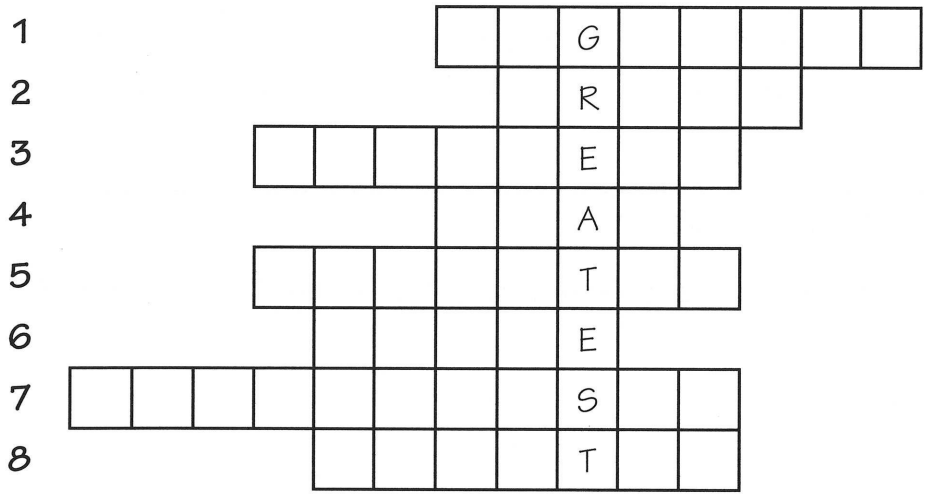
6 mostly disposes of?_____

7 is mostly a head part?_____

8 is chatty inside?_____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

FILL THE GAPS – Write list words to fit each horizontal row.



A to Z – Write the list words in alphabetical order.

WORDS IN A WORD – Find over 60 words in the word GRADIENT. Each word you find must have at least three letters, no 'er' or 'ing' words or names allowed, and each letter can only be used as often as it appears in 'gradient'.

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

8 H

WORD LIST:

- half
- hectare
- heptagon
- hexagon
- HCF
- horizontal
- hypotenuse
- hour
- hundreds
- hundredths
- hash
- height
- hemisphere
- Hindu-Arabic
- histogram

Use the word list above to complete the following exercises:

SENTENCE JUMBLE – Reorder the words so that each sentence makes sense. The first word is capitalised to help you get started.

1 Highest stands Factor HCF for Common

_____

2 sides hexagon equal A has six regular

_____

3 number We the system Hindu-Arabic use

_____

4 are hectare There square 10 000 in metres a

_____

5 graph called histogram A is column statistical a

_____

6 The longest right-angled is side a hypotenuse the in triangle

_____

7 represents digit the The after point second decimal hundredths

_____

WHICH WORD ...

1 is mostly a boy's name? _____

2 ends with weight? _____

3 is mostly far in the distance? _____

4 contains a label? _____

5 is an anagram of the race? _____

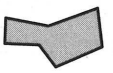
6 mostly belongs to us? _____

7 starts with a spell? _____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	half	600	• LCM
2	heptagon	12, 16 → 4	• hundredths
3	hundreds		• bowl
4	height		• polygon
5	hash	#	• 2
6	horizontal	*h*	• acre
7	hour		• vertical
8	hectare		• base
9	HCF	½	• day
10	hemisphere		• symbol

A to Z – Write the list words in alphabetical order.

_____

_____

_____

PROCEDURE – Complete the following procedure to find the length of a flying fox connected from a 36 metre high tree which stops 105 metres from the tree.

1 A _ _ the s _ _ _ _ _ _ of 36 and _____ .

2 Take the s _ _ _ _ _ r _ _ _ of this t _ _ _ _ to find the l _ _ _ _ _ of the h _ _ _ _ _ _ _ _ _ _ , which is the length of the f _ _ _ _ _ f _ _ .

3 The answer is _____ m _ _ _ _ _ .

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

9 I

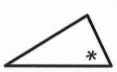
WORD LIST:

- inequality
- infinite
- infinity
- infinitesimal
- integer
- intersect
- irrational
- isosceles
- imperial
- image
- identity
- improper
- index
- indices
- inverse
- interpret
- interior
- irregular
- isometric
- increase
- impossible

Use the word list above to complete the following exercises:

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	inverse	∞	• number
2	infinity	60% like Maths	• original
3	inequality	$\sqrt{2}$	• proper
4	interior	6^4	• parallel
5	irregular	$\frac{11}{5}$	• metric
6	interpret		• certain
7	index	\$5 to \$6	• decrease
8	improper	"	• reciprocal
9	image	0 probability	• exterior
10	irrational		• rational
11	intersect	$\frac{5}{8} \rightarrow \frac{8}{5}$	• exponent
12	integer		• infinitesimal
13	imperial	<	• equation
14	impossible		• regular
15	increase	-9	• analyse

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

TRUE OR FALSE?

1 Isosceles triangles have 2 equal angles.
____

2 -5½ is an integer.____

3 Interior angles of a pentagon add to 540°.____

4 Inequalities have equal signs.____

5 Increasing 50 by 100% = 150.____

6 Parallel lines intersect.____

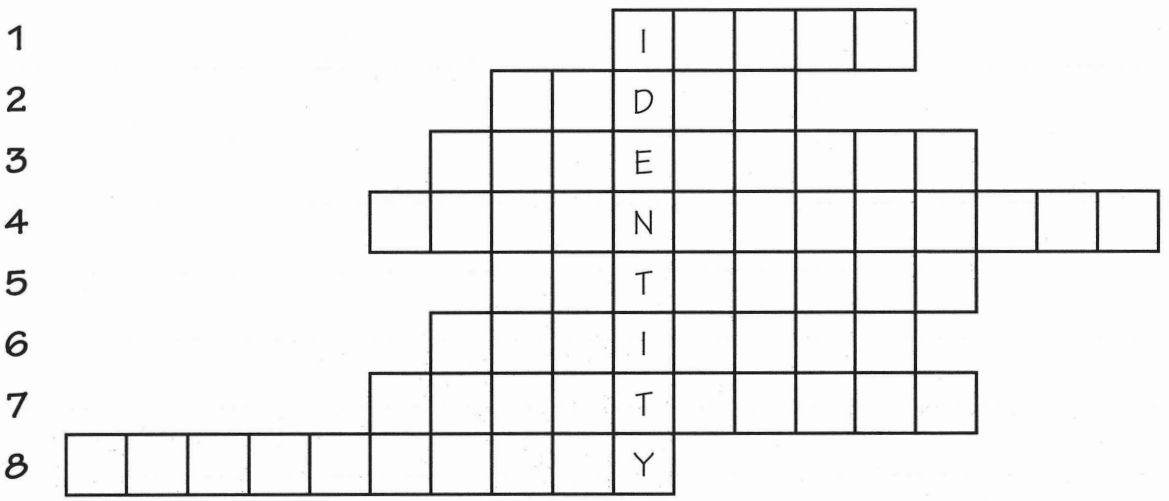
7 1/1 000 000 is infinitesimal.____

8 A smaller image is a reduction.____

9 Improper fractions can't have an inverse.
____

10 Isometric paper shows perspective.____

FILL THE GAPS – Write list words to fit each horizontal row.



DOUBLE MEANING – Write two sentences using each word, one with a mathematical context and the other with a non-mathematical context.

1 inequality_____

_____

2 irrational_____

_____

3 imperial_____

_____

4 index_____

_____

5 improper_____

_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

10 JKXYZ

WORD LIST:

- kite
- kilometre
- kilolitre
- knot
- kilogram
- join
- kilobyte
- x-axis
- y-axis
- year
- zero
- x-intercept
- y-intercept
- key
- justify

Use the word list above to complete the following exercises:

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 kite_____

2 knot_____

3 x-intercept_____

4 kilogram_____

5 join_____

6 justify_____

7 zero_____

8 key_____

SENTENCE JUMBLE – Reorder the words so that each sentence makes sense. The first word is capitalised to help you get started.

1 is The horizontal x-axis

_____

2 kilolitre 1000 There litres are a in

_____

3 another for Key word is legend

_____

4 cuts y-intercept y-axis A the

_____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

5 A adjacent equal pairs kite has sides two of

_____

6 can 1024 a There bytes kilobyte be in

_____

7 the The and y-axis origin x-axis intersect the at

_____

8 a of knot measure A is nautical speed

_____

ABBREVIATIONS – Write the correct abbreviation for each term.

1 kilometre _____

2 kilolitre _____

3 kilogram _____

4 kilobyte _____

5 knot _____

BITS AND PIECES – Use these word parts to form 7 list words.

et	in	li	if	lob	ye	st	ar	il	ept	gr	ki	k
om	ju	rc	y	te	am	tre	yte	kilo	re	o	kil	

1 _____

2 _____

3 _____

4 _____

5 _____

6 _____

7 _____

WHICH WORD ...

1 has wood inside? _____

2 is mostly a head part? _____

3 is mostly a model? _____

4 throws inside? _____

5 begins fairly? _____

6 is mostly negative? _____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

11 L

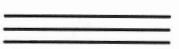
WORD LIST:

- latitude
- longitude
- least
- less than
- length
- line
- linear
- LCM
- litre
- leap year
- lunar month
- label
- law
- like
- likely
- list
- loss

Use the word list above to complete the following exercises:

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	latitude	4 and 6 → 12	• 4 weeks
2	longitude		• greater than
3	less than	366 d	• profit
4	length	<	• quad-annual
5	line	4*ac*, *ac*	• distance
6	LCM		• parallels
7	litre		• unlike
8	lunar month	CP = $50 SP = $40	• plane
9	like		• meridians
10	loss	L	• HCF
11	leap year		• capacity

CLOZE – Use list words to help complete the passage.

Latitude and l _ _ _ _ _ _ _ _ are imaginary l _ _ _ _ on a map or g _ _ _ _ to determine p _ _ _ _ _ _ _.

The lines of l _ _ _ _ _ _ _ are called parallels. The parallel 0° is called the E _ _ _ _ _ _ _. Parallels run east and w _ _ _ and are measured in degrees north and south of 0°. 90° north is the North P _ _ _ and 90° south is the S _ _ _ _ Pole. Latitude is a good indicator of Cl _ _ _ _ _.

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

The lines of longitude are called m _ _ _ _ _ _ _ _ . The meridian 0° is called the Greenwich or P _ _ _ _ Meridian, cutting through a su _ _ _ _ of London. Meridians run n _ _ _ _ and south connecting the 2 poles. Longitude is measured in d _ _ _ _ _ _ east and west of G _ _ _ _ _ _ _ _for 180° and meet at the International D _ _ _ Line in the P _ _ _ _ _ _ Ocean. Longitudes indicate t _ _ _ zones.

Latitude and longitude are actually measured as angles from the c _ _ _ _ _ of the Earth. Though it seems like latitude is along a v _ _ _ _ _ _ _ y-axis and should not be labelled first, l _ _ _ _ _ _ _ is actually a h _ _ _ _ _ _ _ _ _ angle along an x - _ _ _ _ .

HALF THE ALPHABET – Change the bold letter to form a different word. You must use a different letter for each one.

1 l**e**ast_____

2 l**e**ss_____

3 th**a**n_____

4 lin**e**_____

5 lea**p**_____

6 **y**ear_____

7 l**a**w_____

8 mo**n**th_____

9 l**a**bel_____

10 l**i**ke_____

11 l**i**st_____

12 lo**s**s_____

13 l**i**tre_____

TRUE OR FALSE?

1 Higher latitudes have warmer weather. _____

2 Linear graphs are straight lines. _____

3 Lowest and least have the same meaning. _____

4 Like terms can be added. _____

5 LCM stands for Least Common Multiple. _____

6 A line has 2 ends. _____

7 A lunar month is shorter than a calendar month. _____

8 Litres are units of mass. _____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

12 M

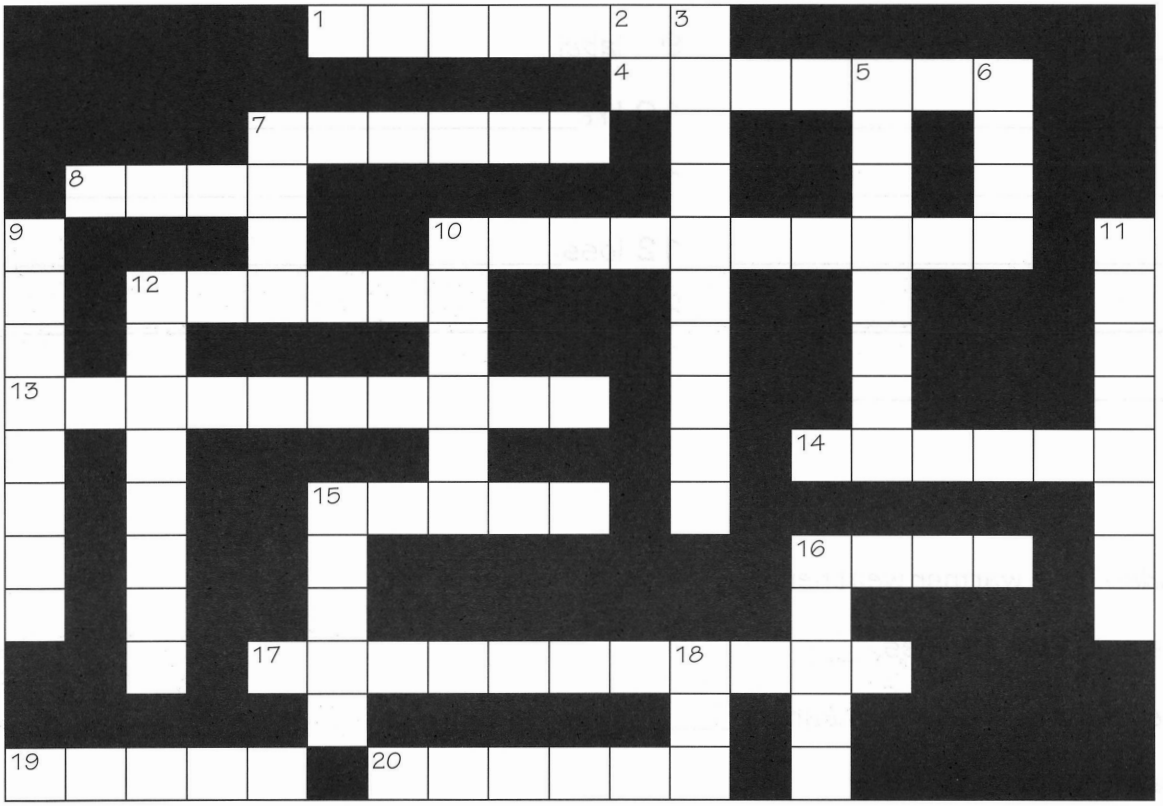
WORD LIST:

- maximum
- minimum
- mean
- median
- mode
- midpoint
- multiple
- multiplication
- minus
- minute
- month
- metre
- millimetre
- millilitre
- mile
- millennium
- millions
- measurement
- metric
- map
- mathematics
- mass
- mnemonic
- money
- mark-up
- method
- magnitude
- mixed
- meridian

Use the word list above to complete the following exercises:

CROSS-FILL

- Use all but five of the list words to complete the crossword (note that mark-up appears without a hyphen and 2 Down is not a list word). Three clues are included to help you get started.
- Then write simple clues for the remaining words.



Across

1 _____

4 _____

7 _____

8 _____

10 _____

12 _____

13 _____

14 _____

15 _____

16 _____

17 _____

19 4 weeks

20 _____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

Down

2 most common last 2 letters of list words

3 mm

5 _____

6 _____

7 _____

9 _____

10 _____

11 _____

12 _____

15 _____

16 _____

18 _____

Which five list words are not in the Cross-Fill?

_____

COMPRENDEZ – Read the following passage and answer the questions below.

In an experiment to see which type of plastic would condense the most water in a desert, the results shown in the table at the right were recorded.

Plastic number	Water (mL)
1	6
2	25
3	17
4	4
5	17
6	15

1 The amounts of water were rounded off to the nearest what?

_____

2 What was the maximum amount of water obtained?

_____

3 Which plastic number condensed the minimum amount of water?

_____

4 What was the mode amount of water?_____

5 What was the mean amount of water?_____

6 What was the median amount of water?_____

7 How is the range calculated? (Use list words in your explanation.)_____

_____

8 What is the range for the water collected?_____

9 What is the total mass of water obtained in this experiment? Explain your answer.

_____

10 Which number plastic would you take on a trip to the desert? Why?_____

_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

13 NQ

WORD LIST:

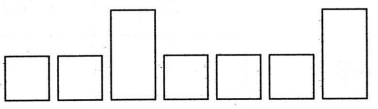
- natural
- number
- numeral
- negative
- nonagon
- normal
- numerator
- nil
- net
- network
- notation
- quadrant
- quadrilateral
- quartile
- quotient
- quarter
- quantity
- question

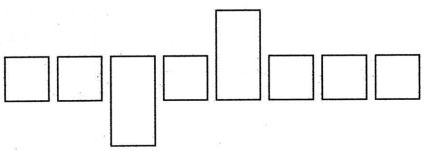
Use the word list above to complete the following exercises:

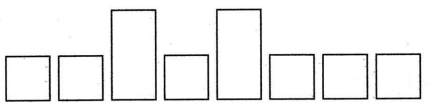
PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

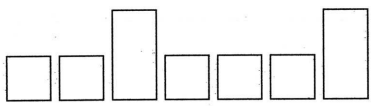
	Noun (naming word)	**Verb (doing word)**	**Adjective (describing word)**
1		question	
2	quantity		
3			quarter
4	notation		
5			normal
6			negative
7		network	

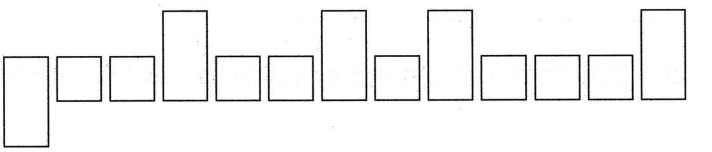
WORD SHAPES – Write list words to fit the word shapes.

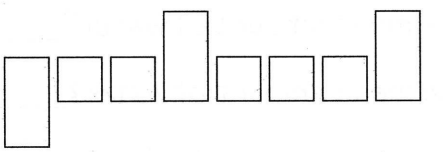
1 

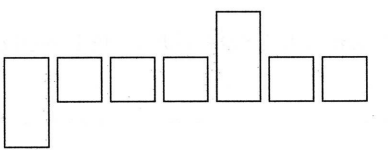
2 

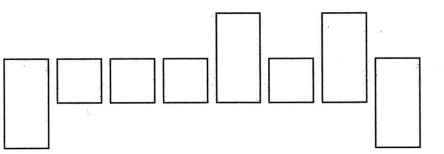
3 

4 

5 

6 

7 

8 

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

FIXES – Write different words (with at least two more letters than given) which begin or end with the following prefixes (on the left) and suffixes (on the right) from list words. You cannot use list words!

1 neg_____ 2 tive_____

3 nat_____ 4 ral_____

5 nor_____ 6 mal_____

7 not_____ 8 ation_____

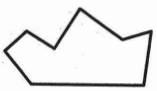
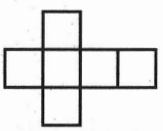
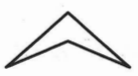
9 net_____ 10 ator_____

11 quad_____ 12 ant_____

13 quar_____ 14 arter_____

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	natural	1, 2, 3, 4	• answer
2	negative		• denominator
3	nonagon	-3	• perpendicular
4	normal	4, **6**, 7, 9, 10	• product
5	numerator		• positive
6	net	÷	• semi-circle
7	quadrant		• median
8	quadrilateral	?	• counting
9	quartile	$\frac{6}{7}$ ←	• square
10	quotient		• polygon
11	question		• solid

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

14 O

WORD LIST:

- oblique
- obtuse
- odd
- octagon
- ordered pair
- origin
- ogive
- operation
- opposite
- open
- oval
- outlier
- ordering
- outcome
- organise
- outline

Use the word list above to complete the following exercises:

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 oblique _____

2 odd _____

3 ordered pair _____

4 ogive _____

5 operation _____

6 opposite _____

7 oval _____

8 outlier _____

TRIOS – Circle the correct term to complete each sentence.

1 An octagon has **6 / 7 / 8** sides.

2 **55° / 155° / 255°** is an obtuse angle.

3 The opposite operation to subtraction is **multiplication / addition / division.**

4 **7 / 8 / 9** is not an odd number.

5 The ordered pair (-3, 6) is in the **first / second / third** quadrant.

6 An ogive uses data from the **cf / fx / f** column.

7 The origin is the centre of the **number line / number plane / number system.**

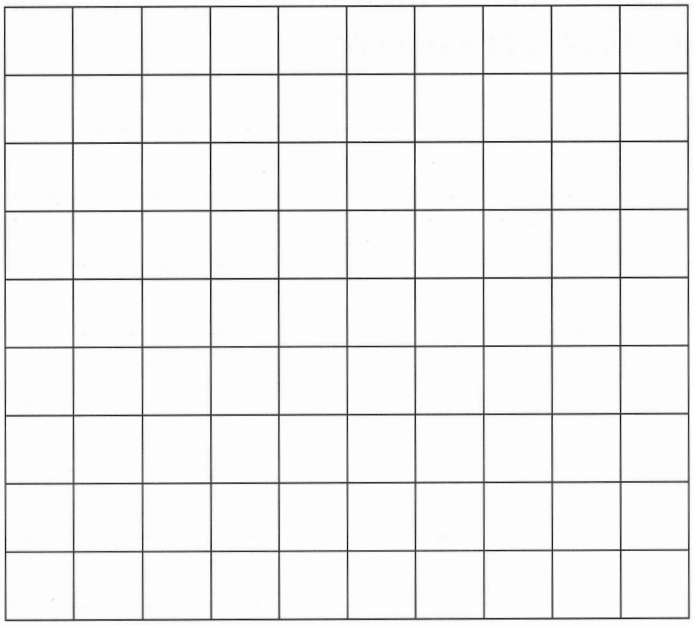
8 An outlier mostly affects the **mode / median / mean**.

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

	Noun (naming word)	Verb (doing word)	Adjective (describing word)
1	operation		
2	origin		
3			open
4			opposite
5		ordering	
6		organise	
7	outline		

AMAZING – Create a word maze by arranging at least 12 list words into the following grid. Then fill in the blank spaces with extra letters and challenge a friend to complete your maze.



WHICH WORD ...

1 ends with jewellery? _____

2 is mostly very light? _____

3 ends with a drink? _____

4 mostly writes? _____

5 mostly hands over? _____

6 tells a fib inside? _____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

15 P

WORD LIST:

- palindrome
- parallel
- perpendicular
- parallelogram
- parentheses
- pentagon
- percentage
- pi
- place
- pie
- perimeter
- plane
- polygon
- power
- prime
- positive
- point
- probability
- proportion
- pyramid
- Pythagoras
- product
- plus
- pattern
- proper
- pronumeral
- perspective
- predict
- prism
- problem
- property
- proof
- protractor
- polyhedron
- possibility
- profit
- plot
- puzzle

Use the word list above to complete the following exercises:

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 palindrome_____

2 pentagon_____

3 Pythagoras_____

4 pyramid_____

5 pattern_____

6 perimeter_____

7 pi_____

8 pie_____

9 property_____

10 power_____

HALF THE ALPHABET – Change the bold letter to form a different word. You must use a different letter for each one.

1 plan**e**_____

2 **p**ower_____

3 pris**m**_____

4 pri**m**e_____

5 plu**s**_____

6 **p**lot_____

7 p**i**_____

8 prope**r**_____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

9 point_____

10 pie_____

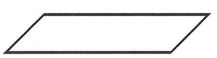
11 puzzle_____

12 place_____

13 parentheses_____

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	parentheses	+	• minus
2	product	%	• composite
3	proper	$\frac{6}{11}$	• brackets
4	pronumeral	p	• quotient
5	plus	×	• ratio
6	proportion	1:3	• possibility
7	probability		• improper
8	plane	29, 17, 5	• skew
9	parallel	( )	• decimal
10	prime		• coefficient
11	percentage	fifty-fifty	• perpendicular

WHICH WORD ...

1 has mostly got it all?_____

2 mostly keeps the rain out?_____

3 robs inside?_____

4 has a ship stop in its centre?_____

5 is mostly frilly?_____

6 starts with mum or dad?_____

7 has the edge in its centre?_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

16 R

WORD LIST:

- radius
- root
- rounding
- radii
- range
- rate
- right
- rhombus
- ratio
- rational
- remainder
- reciprocal
- reflex
- repeating
- regular
- radical
- random
- ray
- rectangle
- reduce
- recurring
- reflection
- rotation
- reduction
- rule
- record
- reason
- relationship
- recognise
- revolution
- represent
- read

Use the word list above to complete the following exercises:

ADJECTIVAL – Adjectives are describing words. Draw lines to join each list word adjective on the left with the most appropriate noun on the right.

1	right	• sign
2	radical	• polygon
3	regular	• decimal
4	repeating	• angle
5	rational	• number
6	reflex	• sample
7	random	• triangle
8	reciprocal	• fractions

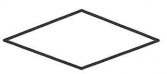
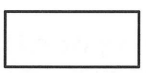
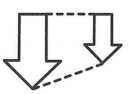
PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

	Noun (naming word)	Verb (doing word)	Adjective (describing word)
1			recurring
2	reflection		
3		reduce	
4		record	
5	relationship		
6			reciprocal
7	remainder		
8		reason	
9		represent	
10		recognise	

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

MATCH AND MIXED?

- Draw lines to match each A word on the left with the most appropriate symbol in the centre and the most appropriate partner word on the right.
- Label each connection as 'S' if the word matches have a similar context or 'D' if the words have a different context; for example, *alternate* and *corresponding* have a similar context, *addition* and *subtraction* have different contexts.

1	root		• square
2	radius	$0.\dot{6}\dot{4}$	• diamond
3	rate	/	• repeating
4	rhombus		• circle
5	ray		• interval
6	rectangle		• diameter
7	reduction	√	• enlargement
8	revolution		• power
9	recurring		• ratio

FIXES – Write different words (with at least two more letters than given) which begin or end with the following prefixes (on the left) and suffixes (on the right) from list words. You cannot use list words!

1 rad_____ 2 tio_____

3 rat_____ 4 duce_____

5 ref_____ 6 ise_____

7 rec_____ 8 ule_____

9 rem_____ 10 ship_____

11 rep_____ 12 cal_____

WHICH WORD ...

1 starts with a truck?_____

2 is mostly an emergency supply?_____

3 is mostly a gift?_____

4 contains a machinery part?_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

17 S

WORD LIST:

- surface
- supplementary
- square root
- similar
- scalene
- sequence
- solid
- square
- subtraction
- sum
- sphere
- set
- second
- spreadsheet
- side
- solve
- sample
- scale
- sector
- semicircle
- segment
- sigma
- sign
- space
- slope
- skew
- simplify
- statistics
- symmetry
- substitution
- standard
- stem-and-leaf
- summary
- straight
- surd
- step
- size
- strategy
- system
- symbol
- sixth
- speed

Use the word list above to complete the following exercises:

PARTS OF SPEECH – Complete the following table by writing the other two forms of each word. Note that some words may be the same.

	Noun (naming word)	Verb (doing word)	Adjective (describing word)
1			square
2	symbol		
3		solve	
4	system		
5			summary
6	second		
7			sample
8	solid		
9			straight
10	strategy		

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 surface_____

2 sequence_____

3 second_____

© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

4 solve _____

5 standard _____

6 step _____

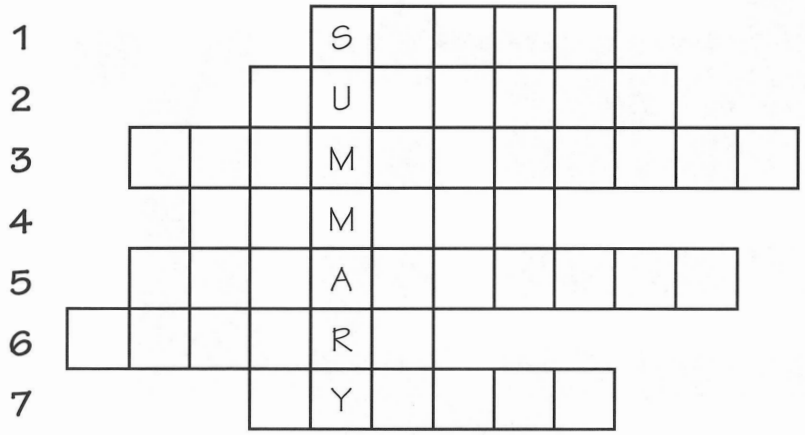
7 speed _____

8 square _____

ADJECTIVAL – Adjectives are describing words. Draw lines to join each list word adjective on the left with the most appropriate noun on the right.

1	supplementary	• form
2	surface	• statistics
3	similar	• edge
4	scalene	• graph
5	solid	• space
6	sample	• lines
7	scale	• plot
8	sector	• area
9	standard	• angles
10	stem-and-leaf	• shapes
11	straight	• object
12	summary	• factor
13	skew	• term
14	sixth	• triangle

FILL THE GAPS – Write list words to fit each horizontal row.



© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

18 T

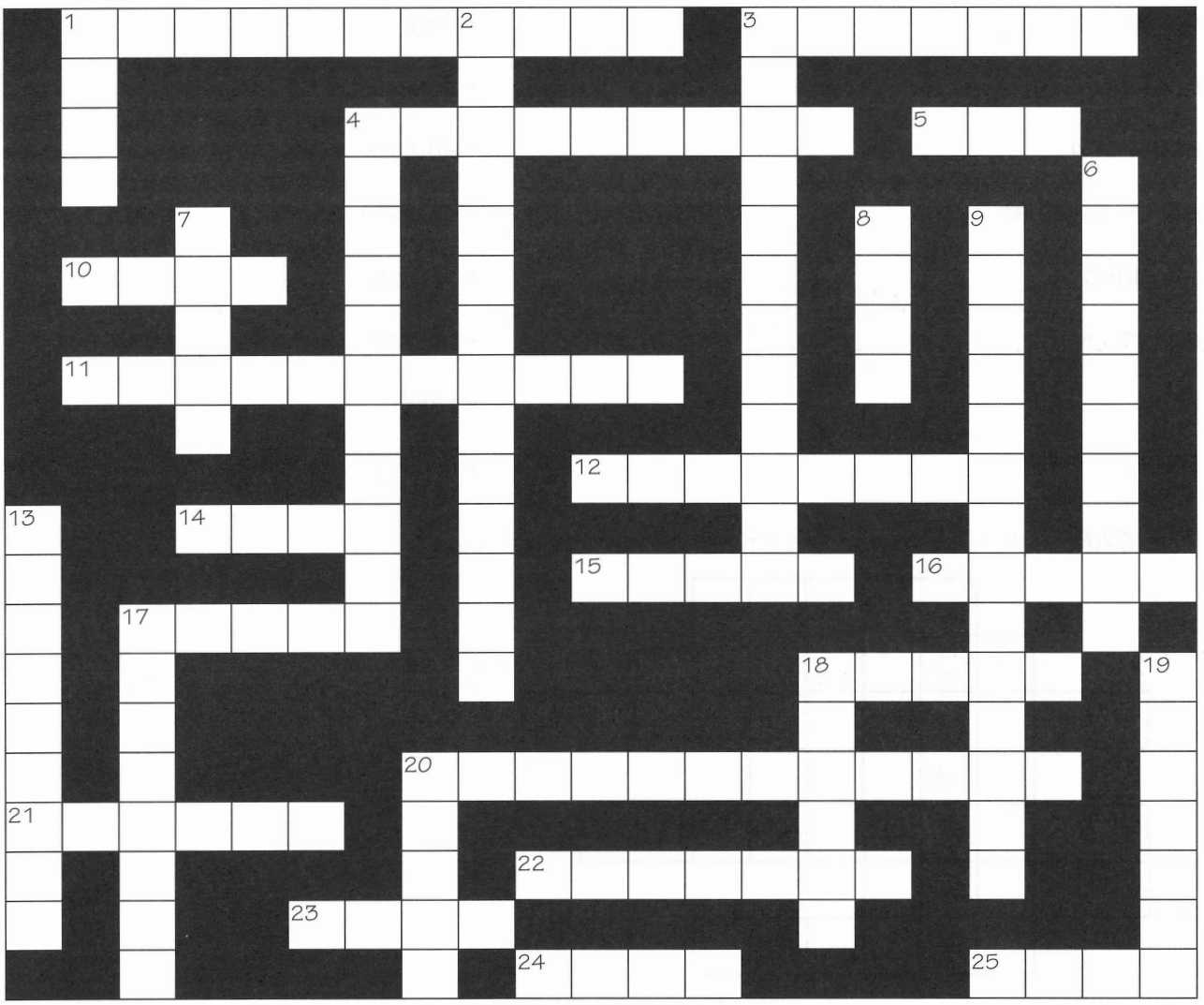
WORD LIST:

- tangent
- trapezium
- triangle
- triangular
- truncated
- total
- tens
- tenths
- thousands
- thousandths
- tonne
- trundle wheel
- tape measure
- true
- term
- translation
- transformation
- take away
- tetrahedron
- theorem
- triad
- transversal
- time
- twenty-four hour
- tessellation
- third
- tax
- table
- travel
- trial
- timeline

Use the word list above to complete the following exercises:

CROSS-FILL

- Use all but three of the list words to complete the crossword.
- Two clues are included to help you get started.
- 'Take away', 'trundle wheel' and 'twenty-four-hour' (without hyphens) appear as single entries.
- 'Tape' and 'measure' and 'time' and 'line' have been split.
- One word appears twice.
- When you have finished, write simple clues for the remaining words.



© Bob Starink and Blake Education 2008
ISBN 978 1 92571 008 3

Across

1 _____
3 _____
4 _____
5 _____
10 _____
11 _____
12 _____
14 _____
15 _____
16 _____
17 _____
18 _____
20 _____
21 _____
22 _____
23 See 24 across.
24 (and 23 across)_____
_____(2 words)
25 (and 19 down)_____
_____(2 words)

Down

1 _____
2 A change to position, orientation or size
3 _____
4 Same as 11 across
6 _____
7 _____
8 Clocks indicate this.
9 _____
13 _____
17 _____
18 _____
19 See 25 across.
20 _____

FIXES – Write different words (with at least two more letters than given) which begin or end with the following prefixes (on the left) and suffixes (on the right) from list words. You cannot use list words!

1 tri_____
2 thou_____
3 tru_____
4 ten_____
5 ter_____
6 tra_____
7 gent_____
8 ndle_____
9 ation_____
10 vel_____
11 iad_____
12 ersal_____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

19 U V W

WORD LIST:

- unit
- variable
- vertical
- vinculum
- volume
- whole
- week
- width
- vertex
- unlike
- unlikely
- unfair
- unitary
- unknown
- valid
- vertices
- value
- weight
- view
- uniform
- vertically
-

Use the word list above to complete the following exercises:

BRANCHING OUT

- List the branches (areas) of mathematics where each word is used. Write N for number, A for algebra, D for data, M for measurement and G for geometry.
- Some words may be used in more than one branch.
- Write a phrase to show how each word is used in each branch; for example, **base** – N, base 10 numbers; M, base × height; G, base of a triangle.

1 unit _____

2 variable _____

3 vertical _____

4 volume _____

5 width _____

6 value _____

7 unfair _____

8 unlike _____

BITS AND PIECES – Use these word parts to form 7 list words.

si ve uni ver ry te ta rti ole lik cul tic un

ia ly var vol vin ume wh es cal ely um ble

1 _____ 5 _____

2 _____ 6 _____

3 _____ 7 _____

4 _____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

WHAT'S THE DIFFERENCE? – What is the difference between:

1 unlike and unlikely? _____

_____

2 vertical and perpendicular? _____

_____

3 weight and volume? _____

_____

4 vertex and vertices? _____

_____

5 a variable and a constant? _____

_____

CHANGE ONE LETTER IN ...

1 **whole** to make an animal. _____

2 **week** to mean look for. _____

3 **unit** to do with wool. _____

4 **valid** to mean dull. _____

5 **weight** to make tall. _____

6 **value** to make an electronic part. _____

NON-MATHS – The passage below uses list words in a non-mathematical context. Use these list words to complete the text: *week, unit, width, view, value, uniform, unlike, valid, whole, unknown, unfair, variable.*

I'm a member of the local cadets. During the last holidays my _____ went for a camping trip that lasted a _____ _____. Proudly wearing my _____, I set up my tent with a _____ of the length and _____ of the campground. The cadet leader asked me which way the wind was coming from. I couldn't name the direction, which was very _____ me. I said the wind was _____ in direction so the answer was _____. He said that wasn't a _____ answer. I thought he was being a bit _____.
He smiled at me and explained how I could work it out. I really _____ his advice and experience.

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

20 Symbols

SYMBOL LIST:

- ≠
- ∞
- ×
- ^
- ≈
- /
- *
- #
- ≤
- %
- ≥
- ”
- ’
- :
- <
- >
- +
- ²
- ⟷
- ⟶
- ——
- ÷
- °
- ∠
- √
- =
- cm
- m
- km
- mm
- ( )
- [ ]
- {}
- g
- kg
- t
- @
- mL
- L
- kL
- ⊥
- ∴
- –
- ³
- ⟶
- ⟶
- $
- ¢
- ∑

Use the word list above to complete the following exercises:

MEANING MATCH – Write each symbol from the list beside its meaning:

1 plus____
2 centimetre____
3 greater than____
4 ray____
5 less than or equal to____
6 angle____
7 kilometre____
8 parentheses____
9 gram____
10 parallel lines____
11 kilolitre____
12 square root____
13 multiply____
14 greater than or equal to ____
15 subtract____
16 cents____
17 brackets____
18 metre____
19 line____
20 per cent____
21 to____
22 millilitre____
23 per____
24 tonne____
25 squared____
26 at____
27 to the power of (caret) ____
28 asterisk____
29 braces____
30 equals____
31 perpendicular to____
32 degrees____
33 minutes____
34 litre____
35 divide____
36 less than____
37 interval____
38 seconds____
39 millimetre____
40 therefore____
41 number (hash)____
42 approximately equal to ____
43 infinity____
44 not equal to____
45 cubed____
46 dollars____
47 sum of (sigma)____

OPPOSITES – Write the symbol opposite to each one shown.

1 + ____
2 = ____
3 ≤ ____
4 —— ____
5 > ____
6 × ____
7 ⊥ ____
8 ² ____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

WHAT'S THE DIFFERENCE? – What is the difference between:

1 > and ≥?

2 = and ≈?

3 kg and L?

4 ⊥ and $\overrightarrow{\quad}$?

5 $\longleftrightarrow$ and $\longrightarrow$?

6 ^{2} and doubled?

7 × and *?

8 [ ] and ( )?

SYMBOLIC – Write the correct symbol to make each expression true. No symbols are repeated.

1 1 cm ____ 9 mm

2 1 kg ____ 1 t

3 24 ____ 8 = 32

4 24 ____ 8 = 3

5 1 cm^3 ____ 1 mL

6 4^2 ____ 8

7 10 ____ 5 = 5

8 60′ = 1 ____

9 1 m^3 = 1 ____

10 10 ____ 5 = 50

11 ____ 9 = 3

12 ____ lines meet at 90°

13 2 ____ 3 = 8

14 60 km ____ hr

15 1.9999 ____ 2

16 100 ____ = $1

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

21 Maths alphabets

ALGEBRA ALPHABET – Write a word used in **algebra** that begins with each of these letters.

A _____

B _____

C _____

D _____

E _____

F _____

G _____

H _____

I _____

J,K,X,Y or Z _____

L _____

M _____

O _____

P _____

R _____

S _____

T _____

U,V or W _____

NUMBER ALPHABET – Write a word used in **number** that begins with each of these letters.

A _____

B _____

C _____

D _____

E _____

F _____

G _____

H _____

I _____

J,K,X,Y or Z _____

L _____

M _____

N, O or Q _____

P _____

R _____

S _____

T _____

U,V or W _____

GEOMETRY ALPHABET – Write a word used in **geometry** that begins with each of these letters.

A _____

B _____

C _____

D _____

E _____

F _____

G _____

H _____

I _____

J,K,X,Y or Z _____

L _____

M _____

N, O or Q _____

P _____

R _____

S _____

T _____

U,V or W _____

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

MEASUREMENT ALPHABET – Write a word used in **measurement** that begins with each of these letters.

A _____ J,K,X,Y or Z _____

B _____ L _____

C _____ M _____

D _____ N or Q _____

E _____ P _____

F _____ R _____

G _____ S _____

H _____ T _____

I _____ U,V or W _____

DATA ALPHABET – Write a word used in **data** that begins with each of these letters.

A _____ J,K,X,Y or Z _____

B _____ L _____

C _____ M _____

D _____ N, O or Q _____

E _____ P _____

F _____ R _____

G _____ S _____

H _____ T _____

I _____ U,V or W _____

SYMBOLS ALPHABET – Write a word used in **symbols** that begins with each of these letters.

A _____ J,K,X,Y or Z _____

B _____ L _____

C _____ M _____

D _____ N, O or Q _____

E _____ P _____

G _____ R _____

H _____ S _____

I _____ T _____

© Bob Starink and Blake Education 2008

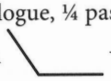
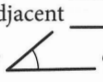
ISBN 978 1 92571 008 3

ANSWERS

1 A

Parts of speech: **1** approximation, approximate. **2** application, applied. **3** add, added. **4** ascent, ascend. **5** analysis, analytic. **6** association, associate. **7** abbreviate, abbreviated.

Branching out: (examples) **1** G (acute angle). **2** N (coordinate axes), D (label the graph axes). **3** N (long division algorithm). **4** A. **5** G (arm of an angle), M (arm of balance beam). **6** M (analogue time). **7** N (addition is a sum), M (addition to find perimeter), A (addition of like terms). **8** N (average run rate), D (average is the mean). **9** M (area of a square is length squared). **10** N (writing integers in ascending order). **11** N (accuracy in arithmetic), M (accuracy in measuring), D (accuracy of a survey), G (accuracy in constructing an angle).

Match and mixed?: **1** alternate corresponding (S). **2** analogue, ¼ past, digital (D). **3** adjacent opposite (D). **4** arm  vertex (S). **5** acute  obtuse (D). **6** addition, +, subtraction (D). **7** approximate, ≈, exact (D). **8** average, $(6 + 8) \div 2$, mean (S). **9** associative, $4 \times (3 \times 5) = (4 \times 3) \times 5$, distributive (S). **10** algorithm, $P = 4L$, rule (S). **11** arithmetic, 7×4, operations (S). **12** analyse, A 60% B 30% C10%, interpret (S). **13** area perimeter (D). **14** axis plane (S). **15** algebra, $6p$, pattern (S).

Which word ...: **1** arc. **2** analogue. **3** associative. **4** acute. **5** average. **6** approximate. **7** ascending.

2 B

Double meaning: (examples) **1** balance an equation, balance on a beam. **2** base 10 numbers, return to base. **3** a bit of data, a bit of pie. **4** work out brackets first, brackets holding up a shelf. **5** bar graph, drink at the bar. **6** braces are grouping symbols, braces hold up your pants.

Words in a word: (examples) thread, bread, bred, read, head, hate, hare, are, bead, dear, dare, rat, rate, tare, tear, date, bat, bate, tea, eat, beat, drab, tar, art, hat, the, tread, hard, bard, red, bed, hear, bear, herb, death, dearth, heart, debt, bet, herd, berth, earth, bare, hare, bath.

Half the alphabet: (examples) **1** c. **2** m. **3** n. **4** k. **5** i. **6** m. **7** d. **8** u. **9** g. **10** v. **11** t. **12** s. **13** f.

Meaning match: **1** 8 bits. **2** binary digit. **3** decimal. **4** cut in half. **5** sphere. **6** reversing. **7** key. **8** unfair. **9** column. **10** construct. **11** edge. **12** width.

Branching out: (with examples) **1** D (biased survey). **2** D (bar graph), N (fraction bar). **3** N (brackets in order of operations), A (expanding brackets). **4** M (balance beam), A (balance an equation). **5** M (boundary of a figure). **6** A (backtracking an equation).

3 C

A dozen please: words applicable – complementary, cone, circle, chord, circumference, congruent, common, cube, cylinder, centre, concurrent, collinear, curve, construct, classification, closed, cone, convex, compasses, cross-section.

Trios: **1** volume. **2** class. **3** counting. **4** coefficient. **5** hat. **6** fractions. **7** convex. **8** 1/100. **9** statistics.

Word shapes: **1** cancelling. **2** coordinates. **3** character. **4** capacity. **5** class. **6** calculator. **7** century. **8** coefficient. **9** certain.

Which word ...: **1** closed. **2** cubic / cube. **3** concave. **4** class. **5** cardinal. **6** cancelling. **7** concurrent. **8** centimetre.

4 D

Branching out: (with examples) **1** G (draw an angle), M(draw a diagram). **2** M (metric system is a decimal one), N (round off to 2 decimal places). **3** G (a decagon has 10 sides). **4** G (diameter cuts a circle in half), M (calculating the diameter). **5** M (decade has 10 years). **6** D (discrete data). **7** N (a discount in price). **8** D (frequency distribution). **9** N (add the distance), M (distance is speed by time), G radius is distance from centre to circumference). **10** D (survey data).

Sentence jumble: **1** A discount is a decrease in price. **2** Integers are directed numbers. **3** Digital time is 12 hour time. **4** A definition is an accurate description. **5** A rhombus can be described as a diamond.

Symbolic: **1** diagonal. **2** digit. **3** dimensions. **4** denominator. **5** descending. **6** divided bar. **7** dozen. **8** dividend, divisor. **9** degrees. **10** diamond. **11** depth.

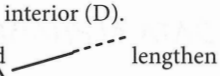
Word shapes: **1** distributive. **2** dodecahedron. **3** discrete. **4** database. **5** display. **6** discount.

5 E

It's a Venn: **Geometry** – ellipse, equiangular, Euclid, edge, exterior. **Algebra** – equation, exponent, expand, expression. **Common** – enlargement, equal, Euler, equivalent, evaluate.

Fill the gaps: **1** enlargement. **2** exterior. **3** exponent. **4** evaluate. **5** equation. **6** eighth. **7** event.

Parts of speech: **1** estimation, estimated. **2** enlarge, enlarged. **3** expansion, expansive (or expanded). **4** equate, equated. **5** explanation, explanatory. **6** experiment, experimental. **7** express, expressive.

Match and mixed?: **1** even, 2, 4, 6..., odd (D). **2** estimate, $49 \times 20 \approx 1000$, evaluate (D). **3** exterior interior (D). **4** enlargement reduction (D). **5** extend  lengthen (S). **6** equal, =, equivalent (S). **7** equilateral equiangular (S). **8** expand, $6(p + 2)$, factorise (D). **9** ellipse oval (S). **10** exponent, u^4, power (S). **11** error, ±, mistake (S).

6 F

Match and mixed?: **1** finite , infinite (D). **2** formula, $P = 4L$, rule (S). **3** frequency, f, number of (S). **4** finance, $, money (S). **5** false, 7 = 8, true(D). **6** fair, fifty-fifty, biased (D). **7** face side (S). **8** fifth, 1/5, five (D). **9** flat level (S). **10** Fibonacci; 5, 8, 13, 21; sequence (S). **11** figurate; 1, 3, 6, 10; triangular (S). **12** factor, $(x + 6)$, product (D).

Non-maths: (In order) Fibonacci, frequency, fortnight, function, finite, fraction, false, factor, flat, fair, face.

Parts of speech: **1** finance, financial. **2** frequent, frequent. **3** falsehood, falsify. **4** factor, factorised. **5** face, face. **6** flat, flatten. **7** formulate, formulaic.

Every third letter: A fractal has all its parts similar to its original shape.

Fixes: (examples) **1** factory. **2** radiance. **3** fracas. **4** delight. **5** freedom. **6** delinquency. **7** fundamental. **8** portal. **9** falter. **10** interaction. **11** finish. **12** surprise.

7 G

Trios: **1** 1 000 000 000. **2** slope. **3** >. **4** braces. **5** before tax. **6** algebra

Write ...: **1** largest, biggest, highest. **2** kg, t, kilotonne. **3** lattice, network. **4** megabyte, kilobyte, byte. **5** vertical rise over horizontal run. **6** histogram, sector, column, line, travel, step, dot, etc. **7** disgusting, 12 dozen.

Which word ...: **1** gross. **2** greatest. **3** graph. **4** greater than. **5** gram. **6** grid. **7** globe. **8** gigabyte.

Fill the gaps: **1** gigabyte. **2** graph. **3** greatest. **4** gram. **5** geometry. **6** globe. **7** generalised. **8** greater.

A to Z: generalised, geometry, gigabyte, globe, gradient, gram, graph, greater than, greatest, grid, gross, grouping.

Words in a word: examples – tag, rig, dig, art, dart, rite, tire, ire, rit, grit, gait, gate, date, raid, grade, gain, rain, near, dear, ear, dare, are, air, dire, ride, eat, tar, rat, tare, rate, grate, great, neat, dent, gent, gin, trade, train, grain, drain, diet, dirge, rent, ten, die, tie, grid, rag, rage, drag, diner, dine, din, den, tan, ran, trade.

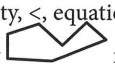
© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

8 H

Sentence jumble:
1 HCF stands for Highest Common Factor. **2** A regular hexagon has six equal sides. **3** We use the Hindu-Arabic number system. **4** There are 10 000 square metres in a hectare. **5** A statistical column graph is called a histogram. **6** The longest side in a right-angled triangle is the hypotenuse. **7** The second digit after the decimal point represents hundredths.
Which word...: **1** half. **2** histogram. **3** horizontal. **4** heptagon. **5** hectare. **6** hour. **7** hexagon.
Match and mixed?: **1** half, ½, 2 (D). **2** heptagon , polygon (S). **3** hundreds, 600, hundredths (D). **4** height, option 8, base (D). **5** hash, #, symbol (S). **6** horizontal ⟷ vertical (D). **7** hour, h, day (S). **8** hectare □ acre (S). **9** HCF; 12, 16 → 4, LCM (D). **10** hemisphere ◡ bowl (S).
A to Z: half, hash, HCF, hectare, height, hemisphere, heptagon, hexagon, Hindu-Arabic, histogram, horizontal, hour, hundredths, hundreds, hypotenuse.
Procedure:
1 add, squares, 105. **2** square root, total, length, hypotenuse, flying fox. **3** 111 metres.

9 I

Match and mixed?: **1** inverse, $\frac{5}{8} \rightarrow \frac{8}{5}$, reciprocal (S). **2** infinity, ∞, infinitesimal (D). **3** inequality, <, equation (D). **4** interior, , exterior (D). **5** irregular  regular (D). **6** interpret, 60% like Maths, analyse (S). **7** index, 6^4, exponent (S). **8** improper, $\frac{11}{5}$, proper (D). **9** image original (D). **10** irrational, $\sqrt{2}$, rational (D). **11** intersect parallel (D). **12** integer, -9, number (S). **13** imperial, ", metric (D). **14** impossible, 0 probability, certain (D). **15** increase, $5 to $6, decrease (D).
True or false?: **1** T. **2** F. **3** T. **4** F. **5** F. **6** F. **7** T. **8** T. **9** F. **10** T.
Fill the gaps: **1** image. **2** index. **3** intersect or irregular. **4** infinitesimal. **5** interior. **6** infinity or infinite. **7** irrational. **8** inequality.
Double meaning: (examples) **1** solve inequality, social inequality. **2** irrational number, irrational behaviour. **3** imperial measures, imperial army. **4** index notation, index finger. **5** improper fraction, improper conduct.

10 J, K, X, Y, Z

Branching out: (with examples) **1** G (a kite is a quadrilateral), M (find area of a kite). **2** M (a knot is a nautical speed). **3** A (*x*-intercept cuts *x*-axis), N (label *x*-intercept). **4** M (1000 grams in a kilogram). **5** G (join vertices). **6** D (justify your interpretation of the data). **7** N (zero is centre of number line), A (raise to the zero power). **8** M (key shows the scale), D (key to a graph).
Sentence jumble: **1** The *x*-axis is horizontal. **2** There are 1000 litres in a kilolitre. **3** Key is another word for legend. **4** A *y*-intercept cuts the *y*-axis. **5** A kite has two pairs of adjacent equal sides. **6** There can be 1 024 bytes in a kilobyte. **7** The *x*-axis and the *y*-axis intersect at the origin. **8** A knot is a nautical measure of speed.
Abbreviations: **1** km. **2** kL. **3** kg. **4** KB. **5** kn.
Bits and pieces: kilobyte, justify, intercept, kilometre, kilogram, kilolitre, year.
Which word...: **1** kilogram. **2** year. **3** kite. **4** kilobyte. **5** justify. **6** knot

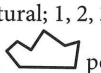
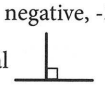
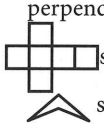
11 L

Match and mixed?: **1** latitude parallels (S). **2** longitude meridians (S). **3** less than, <, greater than (D). **4** length distance (S). **5** line ⟷ plane (D). **6** LCM, 4 and 6 → 12, HCF (D). **7** litre, L, capacity (S). **8** lunar month ☾ 4 weeks (S). **9** like; 4*ac*, *ac*; unlike (D). **10** loss, CP = $50 SP = $40, profit (D). **11** leap year, 366 d, quad-annual (S).
Cloze: (in order) longitude, lines, globe, position, latitude, Equator, west, Pole, South, climate, meridians, Prime, suburb, north, degrees, Greenwich, Date, Pacific, time, centre, vertical, latitude, horizontal, axis.
Half the alphabet: (examples) **1** b. **2** a. **3** e. **4** k. **5** n. **6** g. **7** s. **8** u. **9** i. **10** h. **11** f. **12** t. **13** m.
True or false?: **1** F. **2** T. **3** T. **4** T. **5** F. **6** F. **7** T. **8** F.

12 M

Cross-fill: (example clues in brackets) **Across 1** maximum (most). **4** minimum (least). **7** metric (measurement system). **8** mile (imperial distance) (interchangeable with 6 down). **10** millilitre (mL). **12** minute (unit of time). **13** millennium (1000 years). **14** method (way of doing). **15** money (currency). **16** mass (amount of matter in object). **17** measurement (branch of maths). **19** month (4 weeks). **20** markup (increase in price).
Down 2 um. **3** millimetre. **5** multiple (number divisible by another). **6** mode (most often score) (interchangeable with 8 across). **7** mean (average score). **9** mnemonic (aid to memory). **10** median (middle score). **11** meridian (line of longitude). **12** millions (1000 thousands). **15** mixed (whole and proper fraction). **16** minus (subtract). **18** map (chart).
Missing words: metre, midpoint, multiplication, mathematics, magnitude.
Comprendez: **1** mL. **2** 25 mL. **3** #4 **4** 17 mL. **5** 14 mL. **6** 16 mL. **7** maximum score minus minimum score. **8** 21 mL. **9** 84 grams as each mL has a mass of 1 gram. **10** #2 as it condenses most water.

13 N, Q

Parts of speech: **1** question, questionable. **2** quantify, quantifiable. **3** quarter, quarter. **4** notate, notated. **5** normal, normalise. **6** negation, negate. **7** network, networked.
Word shapes: **1** natural. **2** negative. **3** notation. **4** network. **5** quadrilateral. **6** quotient. **7** quarter. **8** quantity.
Fixes: (examples) **1** negotiate. **2** relative. **3** nation. **4** integral. **5** north. **6** formal. **7** nothing. **8** formation. **9** netting. **10** alligator. **11** quadrangle. **12** depressant. **13** quartz. **14** smarter.
Match and mixed? **1** natural; 1, 2, 3, 4; counting (S). **2** negative, -3, positive (D). **3** nonagon  polygon (S). **4** normal  perpendicular (S). **5** numerator, $\frac{6}{7}$ ←, denominator (D). **6** net  solid (D). **7** quadrant semi-circle (S). **8** quadrilateral square (S). **9** quartile; 4, 6, 7, 9, 10; median (S). **10** quotient, ÷, product (D). **11** question, ?, answer (D).

14 O

Branching out: (with examples) **1** G (oblique lines). **2** N (odd numbers). **3** N (ordered pair on number plane), M (using ordered pairs on maps). **4** D (ogive is cumulative). **5** N (order of operations), A (opposite operations). **6** N (opposite signs), A (opposite operations), M (opposite the right angle), G (opposite angles). **7** G (an oval describes an ellipse). **8** D (an outlier in the data).
Trios: **1** 8. **2** 155°. **3** addition. **4** 8. **5** second. **6** cf. **7** number plane. **8** mean.
Parts of speech: **1** operate, operated or operative. **2** originate, original. **3** open or opening, open. **4** opposition, oppose. **5** order, ordered. **6** organisation, organised. **7** outline, outlined.
Which word ...: **1** ordering. **2** pair. **3** origin. **4** open. **5** ogive. **6** outlier.

15 P

Branching out: (with examples) **1** N (palindromic number) **2** G (pentagon has 5 sides) **3** G (Pythagoras theorem), M (finding

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

hypotenuse using Pythagoras) **4** G (square pyramid), M (volume of a pyramid) **5** A(rule for a pattern), N (number pattern), G (drawing a pattern) **6** G (perimeter of a rectangle) **7** M (pi times diameter), N (pi is irrational) **8** D (pie graph) **9** G (property of angles), N (property of numbers) **10** A (index laws for powers), N (cubing is raising to power of 3).

Half the alphabet: (examples) **1** k. **2** b. **3** e. **4** d. **5** g. **6** z. **7** a. **8** l. **9** j. **10** n. **11** m. **12** t. **13** i.

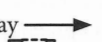
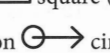
Match and mixed?: **1** parentheses, ( ), brackets (S). **2** product, ×, quotient (D). **3** proper, $\frac{6}{11}$, improper (D). **4** pronumeral, *p*, coefficient (D). **5** plus, +, minus (D). **6** proportion, 1:3, ratio (S). **7** probability, fifty-fifty, possibility (S). **8** plane, skew (D). **9** parallel perpendicular (D). **10** prime; 29, 17, 5; composite (D). **11** percentage, %, decimal (S).

Which word ...: **1** plot. **2** proof. **3** probability. **4** proportion. **5** place. **6** parentheses. **7** prime.

16 R

Adjectival: **1** triangle. **2** sign. **3** polygon. **4** decimal. **5** number. **6** angle. **7** sample. **8** fractions.

Parts of speech: **1** recurrence, recur. **2** reflect, reflective. **3** reduction, reduced. **4** record, record. **5** relate, relative. **6** reciprocal, reciprocate. **7** remain, remaining. **8** reason, reasonable. **9** representation, representative. **10** recognition, recognisable.

Match and mixed: **1** root, √, power (D). **2** radius diameter (S). **3** rate, /, ratio (D). **4** rhombus diamond (S). **5** ray  interval (D). **6** rectangle square (S). **7** reduction enlargement (D). **8** revolution  circle (S). **9** recurring, $0.\dot{6}\dot{4}$ repeating (S).

Fixes: (examples) **1** radiation. **2** patio. **3** ratings. **4** introduce. **5** reference. **6** caramelise. **7** reconcile. **8** module. **9** remember. **10** friendship. **11** repress. **12** conical.

Which word ...: **1** right. **2** rational. **3** represent. **4** recognise.

17 S

Parts of speech: **1** square, square. **2** symbolise, symbolic. **3** solution, solved. **4** systemise, systematic. **5** summary, summarise. **6** second, second. **7** sample, sample. **8** solidify, solid. **9** straight, straighten. **10** strategise, strategic.

Branching out: (with examples) **1** G (surface of a cube), M (surface area). **2** A (rule produces a sequence), N (Fibonacci sequence). **3** M (60 seconds in a minute), N (the second number), A (the second term). **4** A (solve the equation), N (solve the problem). **5** N (standard form), M (standard measure). **6** D (step graph), A (two-step equation). **7** N (speed as an average), M (speed = distance over time). **8** G (square is a special rectangle), N (square number), M (square units).

Adjectival: **1** angles. **2** area. **3** shapes. **4** triangle. **5** object. **6** space. **7** factor. **8** graph. **9** form. **10** plot. **11** edge. **12** statistics. **13** lines. **14** term.

Fill the gaps: **1** sigma. **2** surface. **3** stem and leaf. **4** sample. **5** square root. **6** sphere. **7** symbol

18 T

Cross-fill: (example clues in brackets) **Across** **1** translation (a slide). **3** Tangent (line that touches circle in one point). **4** trapezium (type of quadrilateral). **5** tax (levy). **10** term (combined coefficient and pronumeral). **11** transversal (line that cuts 2 or more others). **12** take away (subtract). **14** tens (column higher than units). **15** table (used for collecting data). **16** total (sum). **17** trial (experimental attempt). **18** third (one out of 3). **20** tessellation (repeating pattern). **21** tenths (column one after the decimal point). **22** theorem (rule). **24** time / line (graph to display historical events). **25** tape / measure (used to measure distance).

Down **1** true (bearing). **2** transformation. **3** trundle wheel (measures distance by turning). **6** triangular (describing a 3 sided object). **7** triad (3 numbers satisfying Pythagoras). **9** twenty-four-hour (method of stating time). **13** truncated (chopped off). **17** triangle (3 sided shape). **18** travel (type of graph). **19** measure. **20** tonne (measure of mass).

Missing words – thousands, thousandths, tetrahedron.

Fixes: (examples) **1** trigger. **2** thought. **3** truth. **4** tension. **5** termite. **6** tradition. **7** intelligent. **8** candle. **9** coronation. **10** gravel. **11** Olympiad. **12** reversal.

19 U, V, W

Branching out: (with examples) **1** M (square units), N (units column). **2** M (variable speeds), A (variables in an equation), D (variables of a graph). **3** G (vertical lines), N (vertical axis). **4** M (volume of a prism). **5** G, M (width of a rectangle). **6** A (value of pronumeral), N (positive value), G (value of an angle). **7** D (unfair survey), N (unfair coin). **8** A (unlike terms), N (unlike signs).

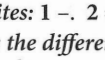
Bits and pieces: unitary, volume, unlikely, vertically, whole, variable, vinculum, vertices.

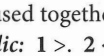
What's the difference?: **1** unlike is algebra term, unlikely a probability term. **2** perpendiculars don't need to be at 90° to the ground. **3** weight units are kg and volume units are L, or weight measures pressure, volume measures space. **4** vertices is plural of vertex. **5** a variable changes, a constant doesn't.

Change one letter: **1** whale. **2** seek. **3** knit. **4** vapid. **5** height. **6** valve.

Non-maths: unit, whole, week, uniform, view, width, unlike, variable, unknown, valid, unfair, value.

20 SYMBOLS

Meaning match: **1** +. **2** cm. **3** >. **4**  . **5** ≤. **6** ∠. **7** km. **8** (). **9** g. **10** . **11** kL. **12** √. **13** ×. **14** ≥. **15** –. **16** ¢. **17** {}. **18** m. **19** . **20** %. **21** :. **22** mL. **23** /. **24** t. **25** ^{2}. **26** @. **27** ^. **28** *. **29** []. **30** =. **31** ⊥. **32** °. **33** '. **34** L. **35** ÷. **36** <. **37** . **38** ". **39** mm. **40** ∴. **41** #. **42** ≈. **43** ∞. **44** ≠. **45** ^{3}. **46** $. **47** Σ.

Opposites: **1** –. **2** ≠. **3** ≥. **4** . **5** <. **6** ÷. **7**  . **8** √.

What's the difference?: **1** > doesn't include the number indicated. **2** ≈ is only approximately equal to, not exactly equal to. **3** kg is a mass unit and L is a capacity unit. **4** ⊥ means perpendicular or at 90° to, the other is parallel. **5** a line has no ends, a ray has one end. **6** squared is to multiply a number by itself, doubling is multiplying it by 2. **7** * is used for multiplication on computers. **8** [ ] go outside ( ) when used together.

Symbolic: **1** >. **2** <. **3** +. **4** ÷. **5** =. **6** ≠. **7** –. **8** ". **9** L. **10** ×. **11** √. **12** ⊥. **13** ^. **14** /. **15** ≈. **16** ¢.

21 MATHS ALPHABETS

Algebra alphabet: Examples – algebra, backtracking, coefficient, distributive, equation, factorise, generalised, hash, inequality, *x*-intercept, like, maximum, opposite, power, root, sequence, term, unlike.

Number alphabet: Examples – addition, billion, cardinal, difference, even, fraction, greater than, hundredths, improper, zero, law, money, natural, operation, palindrome, ratio, sample, total, vinculum.

Geometry alphabet: Examples – arc, ball, complementary, decagon, Euclid, face, geometry, heptagon, image, kite, line, magnitude, net, oblique, parallel, reflex, similar, tangent, vertex.

Measurement alphabet: Examples – area, breadth, capacity, diameter, error, fortnight, gross, hectare, imperial, kilometre, litre, metric, quantity, pi, radius, second, tonne, weight.

Data alphabet: Examples – average, biased, cumulative, data, experiment, fair, graph, histogram, interpret, key, label, mode, quartile, ogive, predict, range, statistics, table, unfair.

Symbols alphabet: Examples – asterisk, brackets, caret, division, equal, greater than, hash, infinity, kilometre, less than, minute, not equal to, parentheses, ray, squared, tonne.

© Bob Starink and Blake Education 2008

ISBN 978 1 92571 008 3

Instant Lessons

Quality ready-to-use resources

Each book in the **Instant Lessons** series is a collection of fresh and versatile worksheet activities which may be photocopied for student use. They are unique in providing **the highest quality learning outcomes with instant usability.**

Versatile uses		Teaching benefits	
Supplementary classroom lessons	✓	Creative teaching ideas	✓
Homework activities	✓	Developing greater student interest and understanding	✓
Reinforcement of learning	✓	A fresh approach to a topic or lesson	✓
Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Literacy in Maths – A–Z of maths terms and ideas

Understanding the language of maths is critical to understanding mathematical concepts. This literacy resource explores the use of mathematical concepts and calculations across the letters of the alphabet. Each topic contains a list of words which are featured in a range of literacy activities for students of all ability levels. The activities provide an emphasis on interpreting, analysing, calculating and appreciating how mathematics is integral to all areas of life.

Topics include

A – angle, apply, appropriate
B – base, button, brackets, braces
C – complementary, chord, classification
D – diamond, distance, dozen
G – geometry, gradient, graph
I – inequality, infinity, isosceles
K – kite, kilometre, kilobyte, key
L – latitude, linear, leap year
O – odd, octagon, organise
R – radius, root, rounding
S – spreadsheet, symmetry, speed
V – variable, vertical, vinculum, volume

Associated titles in the Instant Lessons series

- Literacy in Maths (Books 1, 2, 3, 4)
  - Book 5: Number and algebra
  - Book 6: Data, measurement and geometry
  - Book 7: Maths across the curriculum
- Solving Real-life Stuff with Maths (Books 1, 2)
- Maths In series
- Developing Numeracy series
- Assignments in Junior Algebra (Books 1, 2)
- Assignments in Junior Measurement (Books 1, 2)
- Spreadsheets in Maths (Books 1, 2)

Visit the Blake Education website or email info@blake.com.au for more information about the Instant Lessons series, other mathematics titles, the full list of contents for all titles and ordering advice

Blake EDUCATION
Better ways to learn

Instant Lessons

An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Rd
Clayton South
VIC 3169
+61 3 9558 4433
www.blake.com.au

ISBN: 978-1-92571-008-3